

Rohan Chandra, TS/SCI Clearance

Arlington, VA | 703-627-6422 | rohanchandracpe@gmail.com | www.rohanchandra.com

Professional Summary

Robotics and AI engineer with an active TS/SCI clearance, a Johns Hopkins M.S. in Robotics and Autonomous Systems expected December 2026, and hands-on experience spanning embedded systems, ROS/ROS2, deep learning, and defense-sector software development. Brings a rare combination of perception, planning, and control algorithm expertise alongside production ML engineering and classified mission software — directly aligned with JHU APL's applied research mission.

Education

Johns Hopkins University, Baltimore, MD

M.S. in Robotics and Autonomous Systems, Whiting School of Engineering

September 2023 - Present

Academic GPA: 3.96

- Relevant coursework: Kinematics & Dynamics of Robots, Deep Learning with PyTorch, Reinforcement Learning, Autonomous Vehicles, Introduction to Robotics (TA $\times$ 2 semesters)
- Deep Learning coursework covered CNNs, Vision-Language Models (VLMs), transformer architectures, and multi-modal learning
- Reinforcement Learning coursework covered policy gradient methods, Q-learning, model-based RL, and applied RL to robotic control tasks
- Introduction to Robotics coursework covered PID, Obstacle Avoidance, Sample Based Planning Algorithms, Extended Kalman Filter (EKF), SLAM, and related control algorithms
- Embedded & Autonomous Systems project: implemented a real-time autonomous drone pipeline using Raspberry Pi, RTOS, Arduino, and embedded sensor fusion

University of Virginia, Charlottesville, VA

B.S. in Computer Engineering, School of Engineering and Applied Science

August 2018 - May 2022

Academic GPA: 3.673

Graduated with High Distinction, Minor in Business

- Relevant Coursework: F1/10 Autonomous Vehicles, Embedded Systems and Robotics
- F1/10 Autonomous Vehicles: implemented PID, Follow-the-Gap, and SLAM in ROS/Python on a simulated robot in RViz
- Embedded Systems & Robotics: assembled and programmed a line-following robot using TI-RSLK MAX; designed control flow with Mealy FSM and PWM loop — achieved 2nd-fastest time in class competition

Experience

AI/Machine Learning Engineer, Bana Solutions, Tysons, VA

October 2024 – Present

- Awarded two SPOT performance bonuses; won 1st place in company-wide year-end Hackathon for an original ML concept: Applying VLMs and VQA (Visual Question Answering) to help human analysts efficiently search thousands of images for specific information
- Designed and deployed the Job Role Clustering model, applying graph embeddings, transformer attention mechanisms, and deep neural architectures drawn directly from graduate coursework
- Optimized a graph-based Variational Autoencoder (VAE) with attention for big data processing using Apache Spark and PyArrow, reducing model training time by 20x.
- Researched and prototyped graph attention mechanisms (TGAT, GSAT) as candidate improvements to production models; findings informed the team's future model development roadmap.
- Orchestrated ML workflows using Apache Airflow; tracked model artifacts and deployment readiness with MLflow; monitored data and model drift with Evidently
- Delivered biweekly sprint demos to government program managers and technical leads, increasing transparency and reinforcing confidence in team deliverables

Associate Software Engineer, Northrop Grumman, Dulles, VA

September 2022 – October 2024

- Delivered mission-critical algorithm improvements using Python and C++ for classified satellite programs using a proprietary satellite tool
- Served as the technical liaison between classified and unclassified engineering teams, coordinating cross-boundary development efforts and ensuring continuity across security domains
- Developed multi-threaded applications and 15+ Dockerized microservices; improved CI/CD pipeline performance in Jenkins, maintained unit test coverage, and operated in an Agile workflow using Jira, Confluence, and Bitbucket

FIRST Robotics Software Coach, Nova Labs, Fairfax VA

July 2022 - Present

- Built Nova Labs' robotics software program from the ground up across 5 FTC teams; students have reached state championships multiple times and now mentor rookie teams
- Founded Robot-Go, an 8-week bootcamp covering robotics programming and software engineering best practices; produced an accompanying YouTube video curriculum
- Coached advanced off-season topics (odometry, sensor fusion) to sharpen teams' competitive edge; trained parent volunteers to provide day-to-day student support

ML/Robotics Research Assistant, Prof. Alemzadeh, University of Virginia, Charlottesville, VA

January 2021 – May 2022

- Developed an image augmentation algorithm (rotations, skews, shears) to more than double the size of a surgical robotics training dataset
- Integrated the augmentation pipeline into Mask R-CNN (MRCNN) to improve semi-autonomous robotic surgery model performance

Projects

Final Project for Kinematics and Dynamics of Robots - MoveIt Robot Arm

- Programmed a UR50 robot arm in ROS2 Jazzy to autonomously pick up a simulated factory block and place it on a rover using MoveIt motion planning

RAG-Powered AI Agent - GSA Federal AI Hackathon

- Placed top-10 finalist; built a Retrieval-Augmented Generation agent to consolidate scattered USDA.gov articles into a natural-language query interface
- Integrated Cohere-powered embeddings and auto-generated source bibliographies to improve user trust and transparency in AI-generated answers
- Interviewed by members of the press; delivered live technical presentation to ~30 stakeholders on NLP techniques, system architecture, and real-world impact potential

Turtlebot4 ROS Learning Platform

- Built a personal ROS2 robotics development platform on a ClearPath Turtlebot4; deployed and configured industry-standard packages (SLAM, autonomous navigation) on physical hardware — self-initiated outside of any coursework

University of Virginia HooHacks Hackathon 2022 - TrueForm

- Built a web app that overlays real-time form-correction feedback on exercise videos using computer vision and pose estimation

Arduino Guitar Tuner

- Implemented a DFT algorithm on an Arduino Mega to analyze microphone audio in real time and identify musical notes for guitar tuning feedback

Skills & Interests

- Robotics & Autonomy: ROS, ROS2, Gazebo, RViz, MoveIt, SLAM, EKF, PID, Velocity Obstacles, Turtlebot4, UR50
- ML & AI: PyTorch, CNNs, Transformers, VLMs, RL, RAG, NLP, Graph Embeddings, MLflow, Evidently, Airflow, Cohere, TensorFlow, OpenCV, Mask R-CNN
- Languages: Python, C++, Java
- Infrastructure & DevOps: Docker, Kubernetes, Jenkins, Nexus, Linux, Atlassian Suite (Jira, Confluence, Bitbucket), Git
- Hardware & Embedded: Raspberry Pi, Arduino, RTOS, Embedded C, PWM, FSM design, TI-RSLK, AutoCAD Inventor (Certified 2015)
- Tools: MATLAB, MySQL, Django, Multisim, Ultiboard, VSCode
- Languages (Human): English (native), Japanese, Hindi, Spanish